



FOCUSONFORCE

Testing

Given a customer scenario, describe and recommend an appropriate testing methodology.

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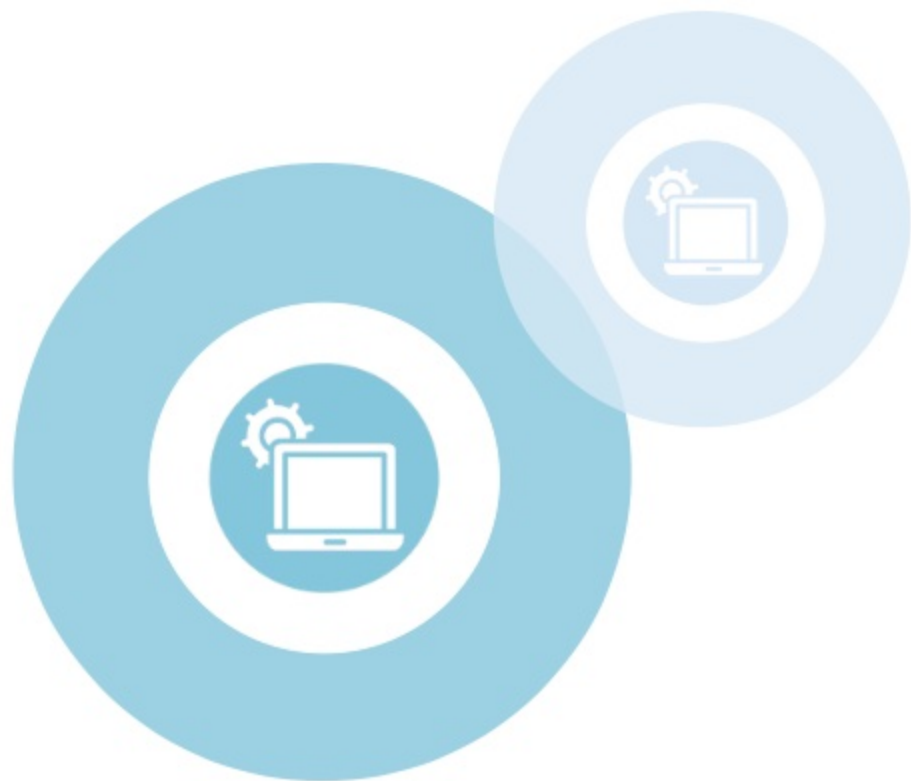
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After studying this topic, you should be able to:

-  Describe the two main categories of testing performed in Salesforce.
-  Explain the different types of testing methodologies utilized in Salesforce.
-  Describe and recommend appropriate testing methodologies for various requirements.

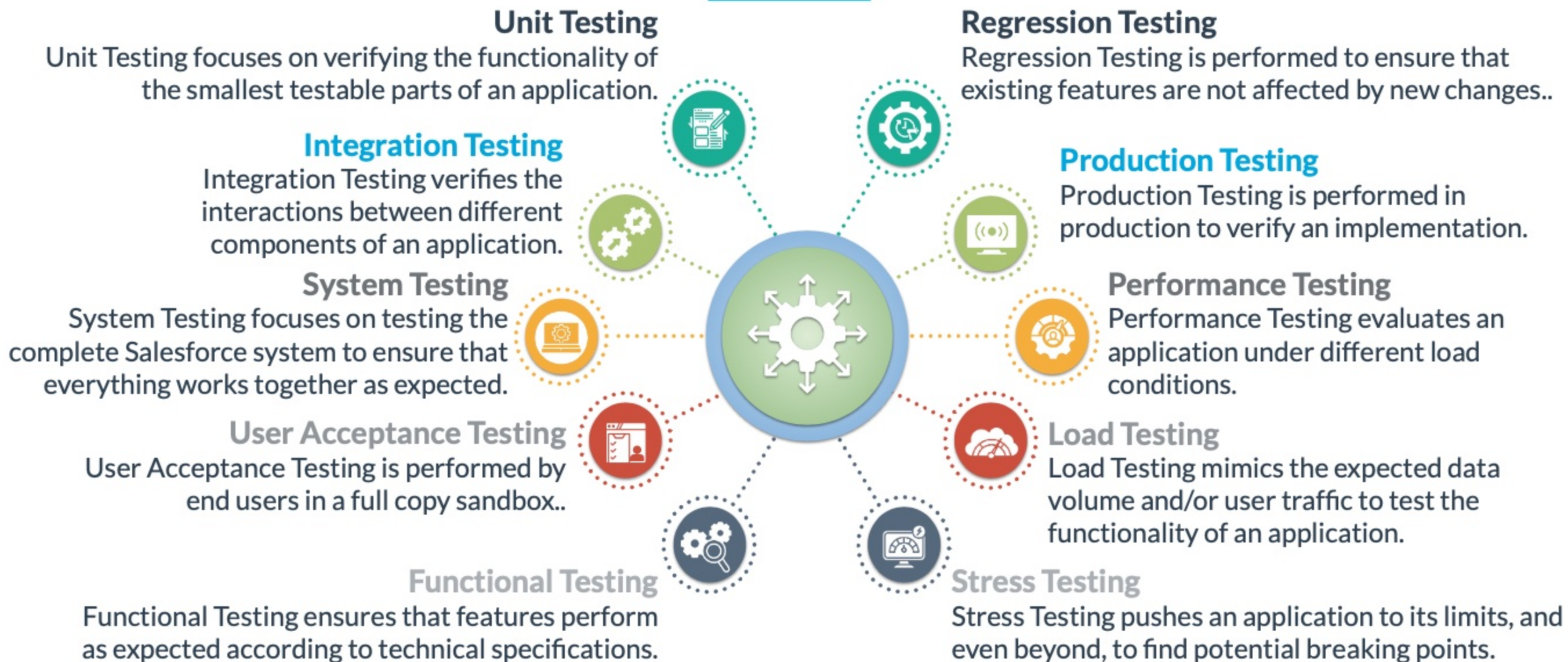


Introduction

In the Salesforce landscape, a comprehensive testing approach is vital for ensuring applications meet both performance and business criteria. This approach includes **unit testing** to check individual components, **integration testing** for verifying component interactions, and **system testing** to assess the application as a whole. **User Acceptance Testing (UAT)** confirms the solution aligns with user expectations and business needs. **Regression testing** ensures new updates do not disrupt existing functionalities. On the performance front, **performance testing**, **load testing**, and **stress testing** evaluate how the application behaves under various loads and conditions. Collectively, these methodologies guarantee that Salesforce applications are robust, reliable, and ready for deployment.



Salesforce Testing Methodologies



Testing Methodologies

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Salesforce Testing Categories

Salesforce testing can be performed **manually** or **automated** using automation tools.

❖ MANUAL TESTING

In manual testing, tests are **set up and executed manually** using traditional methods with the **minimum use of tools**. It can be used for a variety of different test types and testing scenarios. However, it requires performing **repetitive** tasks, **does not scale**, and is prone to **human error**.

❖ AUTOMATED TESTING

Automated testing involves the use of **automation tools** to test Salesforce applications or features. Test scripts are **executed automatically** at scale. It **avoids repetitive and error-prone manual testing activities**. Examples are Selenium, Copado, Provar, ACCELQ, etc.



Salesforce Testing Categories

Salesforce testing can be performed **manually** or **automated** using automation tools.

❖ USING MANUAL TESTING

Manual testing is suitable for testing scenarios that require **human judgment or subjective evaluation**. It can be used to test **one-off, non-repetitive scenarios** that would be counter-productive to automate.

❖ USING AUTOMATED TESTING

Automated testing can be used to test **repeatable test scenarios** that would require **considerable time and manual effort**. It can be used for **regression testing, data-driven testing** with various input values, data sets, and data scenarios, integrating testing into the **CI/CD pipeline**, etc.



Types of Salesforce Testing

Several different **types of Salesforce testing** can be utilized based on the **testing requirements**.



UNIT TESTING

Unit tests focus on verifying the **smallest testable parts** of an application, typically individual **Apex methods**, to ensure that they **work correctly in isolation**.



INTEGRATION TESTING

Integration tests focus on verifying the **interactions between different components**, such as **Apex classes and external APIs**, to ensure that they work as expected.



SYSTEM TESTING

System testing focuses on testing the **complete Salesforce system** to ensure that **all features, configurations, and customizations** work together as expected.

Types of Salesforce Testing

Several different **types of Salesforce testing** can be utilized based on the **testing requirements**.



USER ACCEPTANCE TESTING

User Acceptance Testing (UAT) is **performed by end-users** to ensure that the **system meets business needs and requirements**. It is the final step before deployment.



FUNCTIONAL TESTING

Functional testing is used to ensure that features perform as expected according to technical specifications. Each **function is examined** by **providing input** and **comparing the output** to functional requirements.



REGRESSION TESTING

Regression testing is performed **after changes (new features, updates, enhancements, etc.) to the system** to ensure that existing functionalities are not adversely affected.

Types of Salesforce Testing

Several different **types of Salesforce testing** can be utilized based on the **testing requirements**.



PRODUCTION TESTING

Production testing is performed in the **production environment** to ensure the **successful implementation** of all the code and configurations.



PERFORMANCE TESTING

Performance testing focuses on evaluating the **performance, scalability, and responsiveness** of a Salesforce application under different load conditions.



LOAD & STRESS TESTING

Load testing focuses on mimicking the **expected user traffic** to see how an application performs under that load. **Stress testing** pushes it to its **limits, and even beyond**, to find potential breaking points.

Considerations for Unit Testing

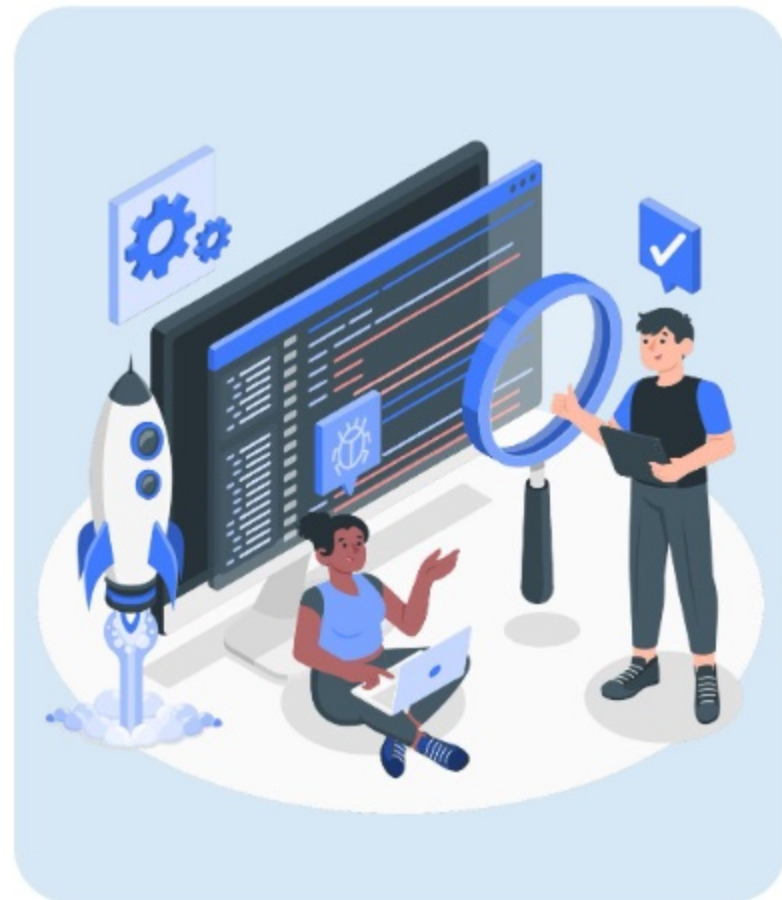
Unit testing involves **testing individual components or units of code** to ensure their correctness and functionality in isolation.

❖ APEX TEST METHODS

Developers use unit testing to verify the behavior of Apex code. **Apex test methods** are usually very specific and can be used to test the behavior of individual **Apex classes**, **triggers**, **Visualforce pages**, or **flows**. **Assertions** can be used to verify the Apex code behavior.

❖ CODE COVERAGE

Salesforce requires a **minimum of 75% code coverage** for Apex code to ensure that most of the code is tested, which is **crucial before deployment to production**.



Considerations for Unit Testing

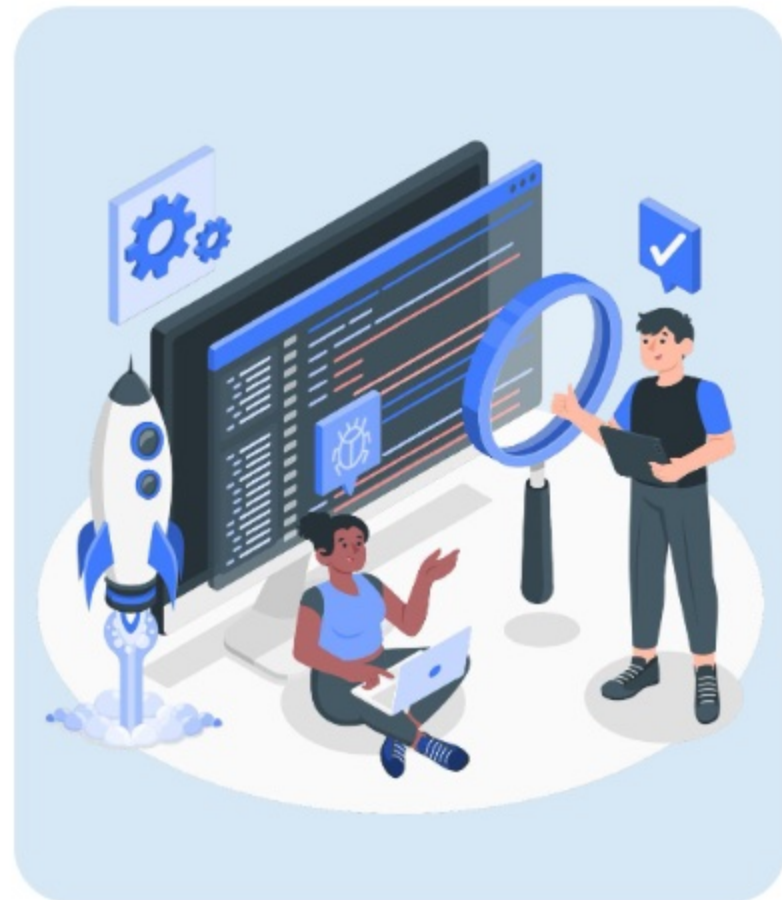
Unit testing involves **testing individual components or units of code** to ensure their correctness and functionality in isolation.

❖ TEST CLASSES & METHODS

Apex test classes and methods use the **@isTest** annotation. Test classes don't count toward the limit of **6 MB of Apex code** in each org.

❖ CONSIDERATIONS

Test methods **cannot send emails or make callouts** to external services. A **mock object** is needed whenever Apex code needs to make an **HTTP callout**. A mock can also be used to **bypass the dependencies of other objects, methods, states, or behaviors**.



Considerations for User Acceptance Testing

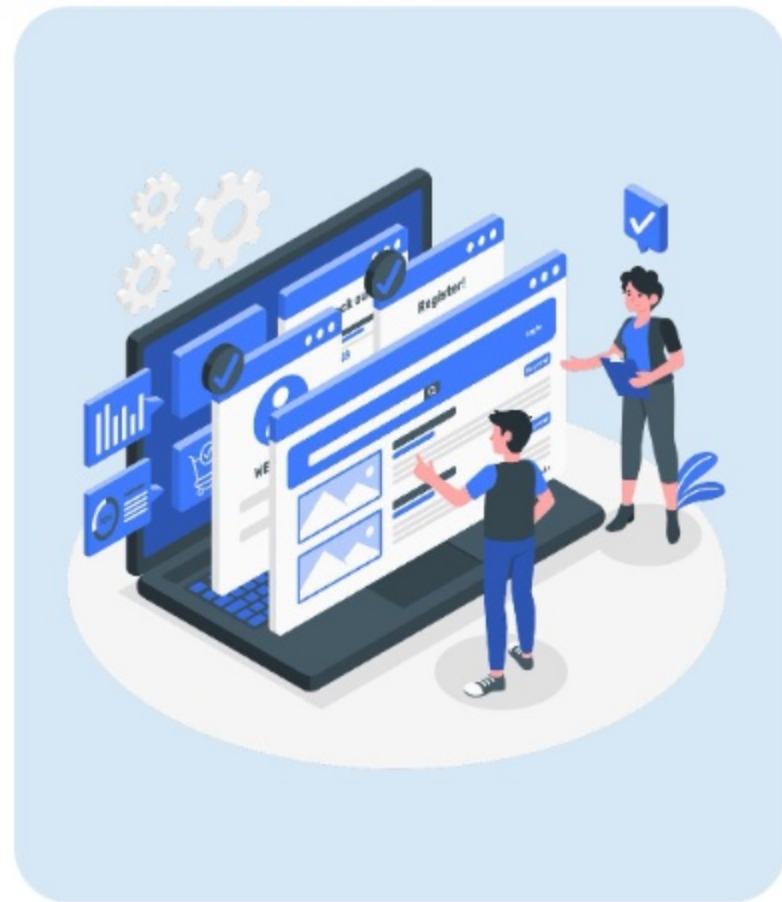
User Acceptance Testing (UAT) is conducted by end-users **before deployment to production** to ensure that an application can **handle the required tasks in real-world scenarios**.

❖ VALIDATION

UAT is critical for validating an application's **usability** and **functionality** from the **user's perspective**, ensuring it aligns with business processes.

❖ RTM

A **Requirements Traceability Matrix (RTM)** can be used to ensure that **all the requirements have test cases** associated with them so that no functionality is left untested during user acceptance testing.



Considerations for Performance Testing

Performance Testing simulates **real-world usage conditions** to identify **potential performance bottlenecks** and **optimize an application's performance** before it affects end-users.

❖ OBJECTIVES

Performance Testing determines **areas of the application** where performance issues might occur, assesses **how well it can scale** with an increasing number of users or data volume, verifies that **it remains stable** under various stress conditions, and **provides insights** into optimization.

❖ APPROVAL

Performance tests must be **submitted for approval** to Salesforce Support **at least two weeks in advance** of the testing date. Unapproved testing is subject to throttling or blocking. The **business case or scenario** and **business justification** must be provided for performance testing.



Considerations for Performance Testing

Performance Testing simulates **real-world usage conditions** to identify **potential performance bottlenecks** and **optimize an application's performance** before it affects end-users.

❖ TEST PLAN

A test plan can be generated and a testing window can be scheduled with the **Performance Assistant**. The test plan can be created based on the **scenarios** that need to be tested and their **success criteria**.

❖ MONITORING

Salesforce Support **does not design or interpret customer performance test results**. Its role is strictly to **monitor test activity** to ensure that there are **no issues with Salesforce services**.



Considerations for Performance Testing

Performance Testing simulates **real-world usage conditions** to identify **potential performance bottlenecks** and **optimize an application's performance** before it affects end-users.

❖ SANDBOX ONLY

Performance testing, including **load** and **stress testing**, can **only be performed in a sandbox**. It is **not permitted in production environments** since Salesforce is a **multi-tenant platform**.

❖ OTHER CONSIDERATIONS

Performance testing must take **governor limits** into account. Built-in monitoring tools, such as the **Developer Console**, **Lightning Usage App**, and **Salesforce Optimizer**, can be used to **monitor** and **analyze performance metrics**.



Considerations for Automated Testing

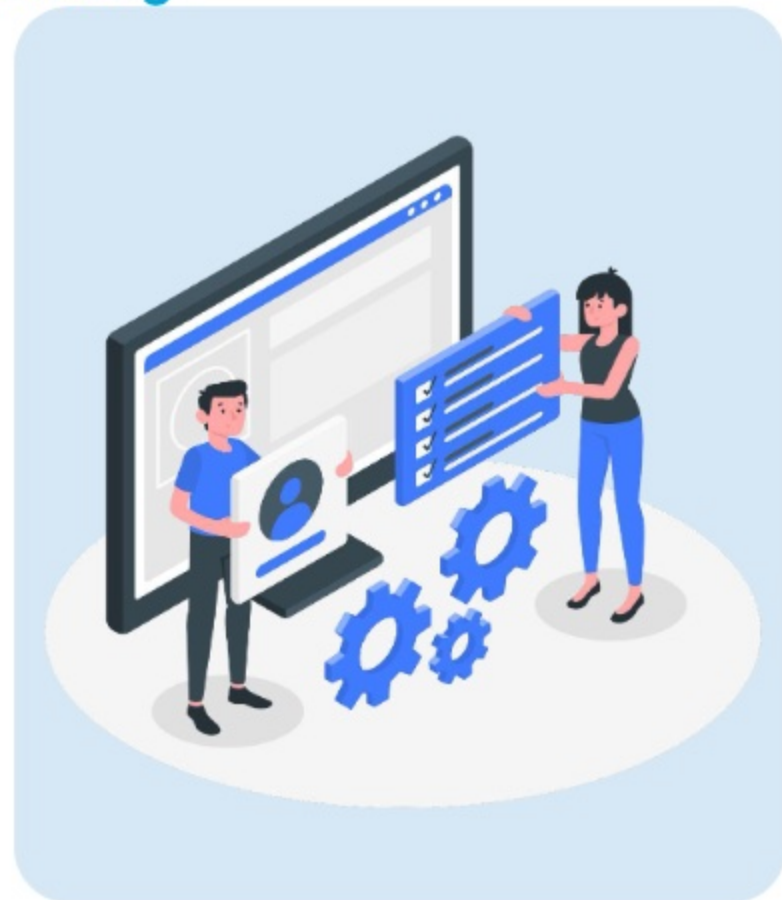
Automated testing tools can be utilized for **unit testing**, **integration testing**, **functional testing**, **regression testing**, **performance testing**, and **UI testing**.

❖ CONTINUOUS INTEGRATION

Test automation can be run for different types of tests, especially **unit and functional tests**, as part of a **continuous integration strategy**. This is useful to **prevent regression**, ensuring that newly released features do not break other previously existing and stable functionality.

❖ MERGING

When using an automated testing tool to help **manage deployments** between Salesforce environments, **automated tests** should be run when **branches are merged**.



Business Scenario

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Business Scenario

Read the business scenario below and determine the most appropriate solutions for the company's requirements.



Cosmic Retail Group, a rapidly growing retail chain, leverages Salesforce for customer engagement and various internal operations. It utilizes Sales Cloud for sales processes, Service Cloud for customer interactions, and various custom applications on the Lightning platform for retail operations. With an expanding footprint and the need for innovative features to enhance its business processes and streamline operations, there's a pressing need for new custom applications in Salesforce.

The development teams have embarked on multiple Salesforce projects for the development of new functionalities, which require robust testing methodologies to ensure efficiency and reliability. The company's Salesforce architect needs to analyze the given technical requirements and recommend appropriate testing methodologies.

Requirements & Solutions

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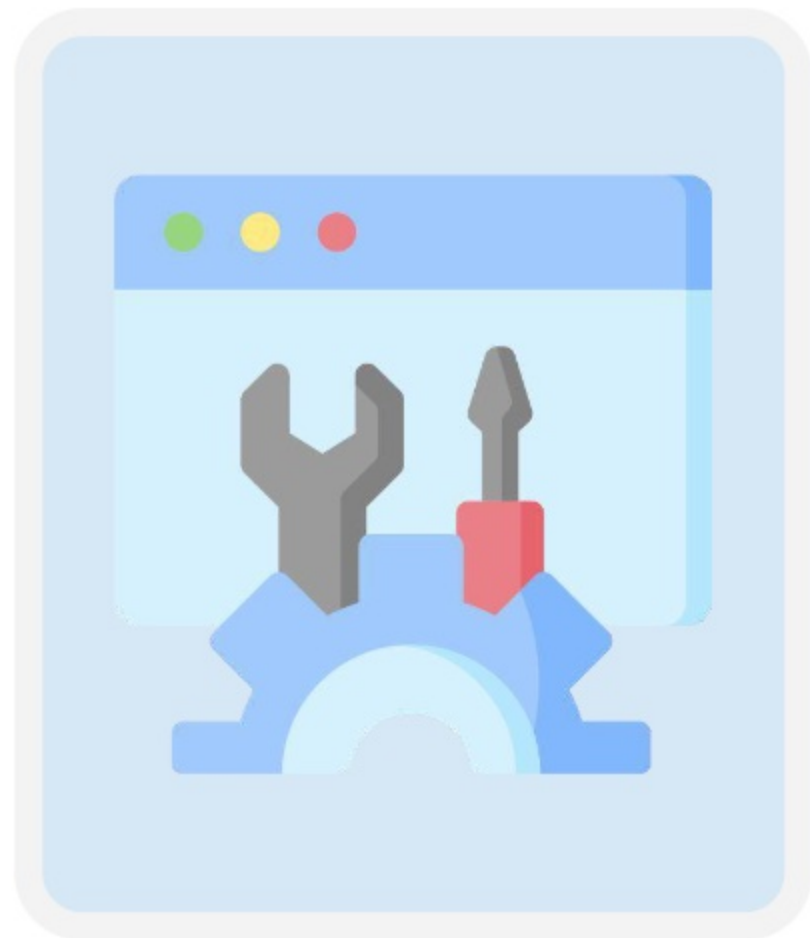
Requirement 1 (Part 1)



SCENARIO

A development team has started working on a custom application for order management in Salesforce. The team has the following technical requirements:

- 1) A custom REST web service has been developed. It processes orders and order line items received from an external system. It is expected to receive more than 150,000 orders per day, and each order can have up to 20 line items. An inbound request can contain only one order and its corresponding line items. The architect needs to recommend testing methodologies to ensure that users don't face platform slowdowns during peak business hours.

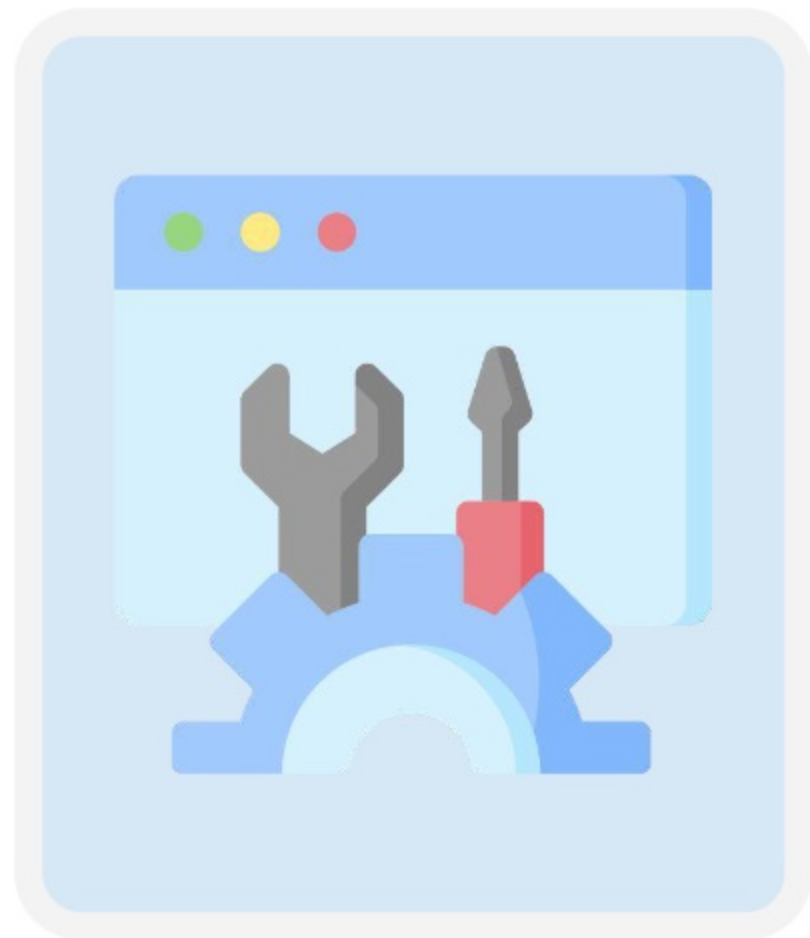


Requirement 1 (Part 2)



SCENARIO

- 2) HTTP callouts are used in Apex code to send processed order information to an external system. The architect must provide relevant considerations for writing Apex test classes for testing HTTP callouts.
- 3) The development team is considering the use of an automated testing tool in a Continuous Integration (CI) workflow to manage deployments between environments. A suitable testing strategy is required to prevent regression.
- 4) When the application components have been deployed in a UAT environment, the team wants to ensure that no functionality is left untested during user acceptance testing.

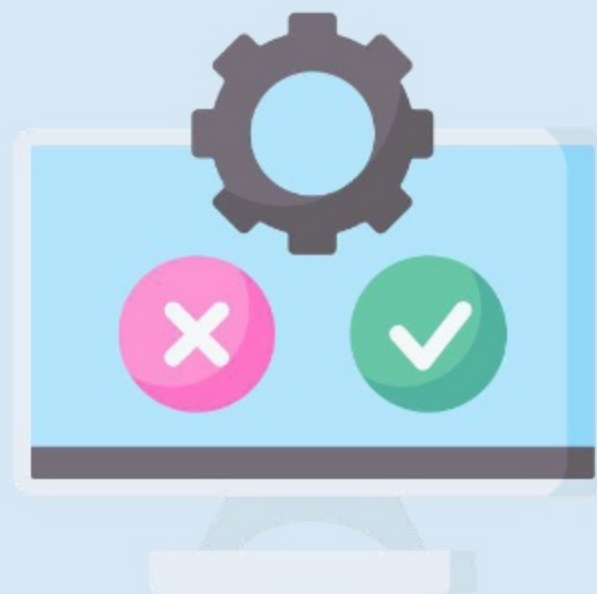


Solution 1 (Part 1)



SOLUTION

- 1) To ensure that users don't experience slowdowns during peak business hours, **performance testing** should be conducted in a full copy sandbox. **Load testing** can be performed using the expected load of 150,000 orders and 20 line items per order.
- 2) To test HTTP callouts in Apex test classes, **mock objects** are required. An **HTTP response mock** can be created by defining a class that extends the **HttpCalloutMock** interface.
- 3) When using an automated testing tool for deployments, **automated tests should be run** when **branches are merged**. **Unit and functional test automation** can be utilized as part of the continuous integration strategy to **prevent regression** and ensure that the new features do not break any existing and stable functionality.

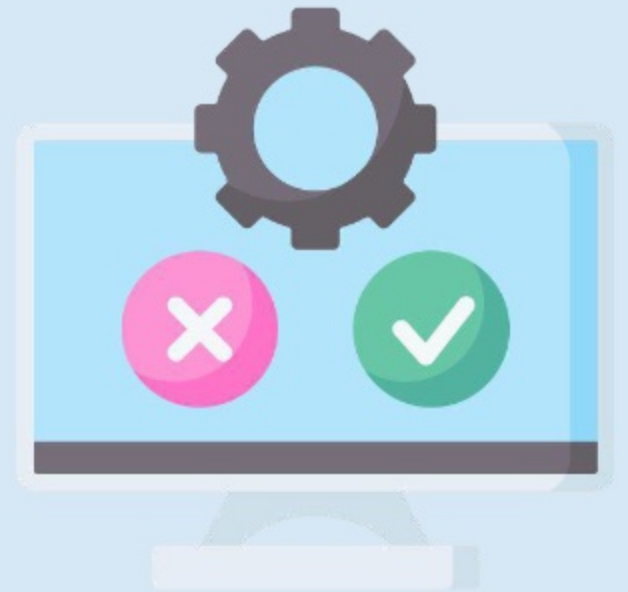


Solution 1 (Part 2)



SOLUTION

- 4) A **Requirements Traceability Matrix (RTM)** document can be used to map user requirements to test cases. It can be used to ensure that **all the requirements have test cases** associated with them and no functionality is left untested during user acceptance testing.

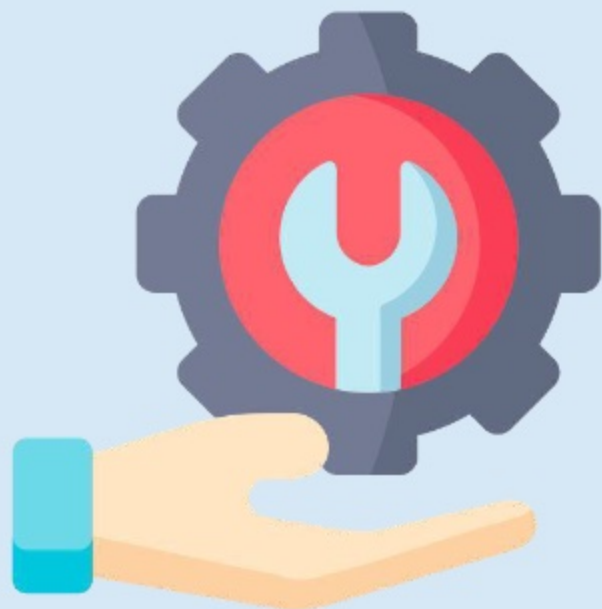


Requirement 2



SCENARIO

A custom application comprising multiple Lightning web components has been created for case management. The components use validation rules and flows defined on the Case object. It will be used by support agents to handle inbound customer service calls. The architect must recommend testing methodologies to ensure that the application does not experience slowdowns under high call volume. Any considerations related to testing must also be described.

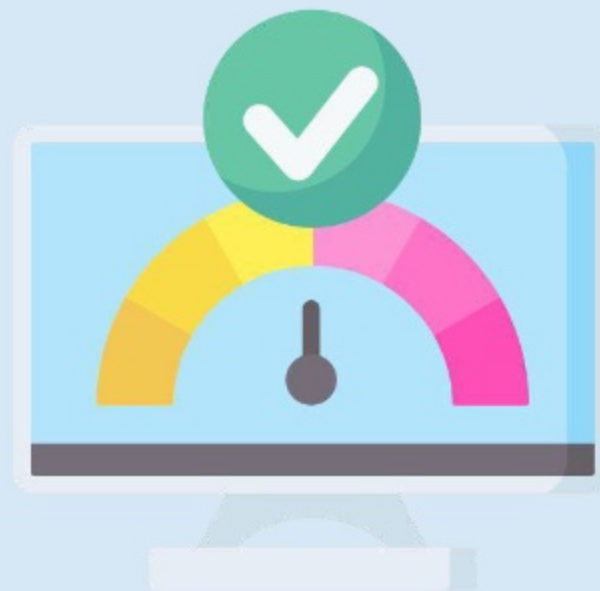


Solution 2



SOLUTION

Performance testing and **load testing** can be performed in a **full copy sandbox** to ensure that the application does not experience slowdowns during high call volume. The request for performance testing must be submitted to Salesforce Support for approval **at least two weeks in advance**. A **business justification** must be provided. Salesforce **monitors test activity** to ensure that there are no issues with Salesforce services. It is important to note that Salesforce **does not permit performance testing in production** since it's a multi-tenant environment with shared resources.



Learn More

- [!\[\]\(467d80e979964f7f8c752fb22248b5b7_img.jpg\) Apex Testing](#)
- [!\[\]\(b71552d33dbf62adf5e5199a70ee02bf_img.jpg\) Advanced Apex Testing](#)
- [!\[\]\(03134b765d1473836ff001925b1b0550_img.jpg\) Types of Salesforce Testing and When To Use Them](#)
- [!\[\]\(aed6947356668967079310026052edc0_img.jpg\) Salesforce Testing](#)
- [!\[\]\(e61aeb0d9066d5d9e54d9b655f50da3d_img.jpg\) Complete Guide to Salesforce Testing](#)
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- [!\[\]\(1d505a46c82c5cefa23b88c2eee900ce_img.jpg\) Continuous Integration](#)

